Md Shohel Rana, Ph.D.

Assistant Professor | School of Computing | College of Engineering and Computing
Georgia Southern University · Updated October 2026

Computing researcher and educator with experience in academic teaching and research in the United States and Bangladesh, alongside earlier software engineering and team leadership roles. My work connects trustworthy AI, cybersecurity, computer vision, and multimodal sensing, with applications in media forensics, computational health, and sustainable agriculture.

Profiles: Google Scholar · ORCID · ResearchGate · Personal website

Education

2017 – 2021	The University of Southern Mississippi Ph.D. in Computational Science (Major: Computer Science) Dissertation: Analyzing and Detecting Android Malware and Deepfake
2013 – 2014	Mawlana Bhashani Science and Technology University M.Sc. (Engr.) in Computer Science and Engineering Thesis: Inpainting on Digital Images using Modified Convolution Method
2006 – 2010	Mawlana Bhashani Science and Technology University B.Sc. (Engr.) in Computer Science and Engineering Thesis: Speckle Noise Reduction in Ultrasound Images using Anisotropic Diffusion Filter

Academic Employment

Aug 2025 – Present	**Assistant Professor**, Georgia Southern University
Jan 2025 – Aug 2025	**Assistant Professor**, University of Massachusetts Dartmouth
Aug 2023 – Jan 2025	**Assistant Professor**, Florida Gulf Coast University
Jun 2022 – May 2023	**Assistant Professor (Teaching)**, University of Alabama at Birmingham
Aug 2021 – May 2022	**Visiting Assistant Professor**, Northern Kentucky University
Aug 2017 – Aug 2021	**Graduate Assistant**, The University of Southern Mississippi
Sep 2016 – Jul 2017	**Senior Lecturer**, Daffodil International University
Jan 2014 – Aug 2016	**Adjunct Faculty**, Daffodil International University

Industrial Employment

Dec 2015 – Sep 2016	**Senior Software Engineer (Associate Lead)**, Orion Informatics Limited
May 2014 – Nov 2015	**Software Engineer (Lead)**, BRAC IT Services Limited
May 2011 – Apr 2014	**Jr. Software Engineer & Software Engineer (Team Lead)**, Sarah Technologies Limited

Honors and Awards

2024	**1-Year Employee Service Award** Awarded by Florida Gulf Coast University, 2024.
2024	**Best Presentation Award**

Awarded by the IEEE International Conference on Information and Computer Technologies, Hawaii, United States, 2024.

2017 – 2021 **Fully Funded Scholarship & GA Award**
Awarded by The University of Southern Mississippi, United States.

2017 – 2018 **Distinguished Researcher Award: Consecutive Award Recipient in 2017 and 2018**
Awarded by Daffodil International University, Dhaka, Bangladesh

2017 **Best Paper & Best Presenter Award**
Awarded by the International Conference on Frontiers of Image Processing, Nepal, 2017.

2004 **Student Award**
Awarded by the Rural Development Academy, Bangladesh.

RESEARCH PROJECTS

2025 – Present **AI-Powered Quantum Sensing and Wearable-Based Non-Invasive Glucose Monitoring System**
Focus: Non-invasive continuous glucose monitoring using quantum-enhanced Raman spectroscopy, wearable physiological sensing, and AI-driven multimodal data fusion for smart healthcare and diabetes management.
Outcomes: Proposed a novel AI-powered quantum sensing framework for non-invasive glucose estimation by integrating Raman spectroscopy, wearable biosignals, and intelligent machine learning models. The system focuses on detecting glucose-related optical and physiological patterns without finger-prick blood sampling. The proposed approach combines quantum-enhanced Raman signal acquisition, wearable sensor data, and AI-based prediction models to support continuous glucose monitoring, personalized diabetes care, and early risk detection. The project also lays the foundation for future smart health applications that involve preventive interventions, uncertainty-aware prediction, and explainable clinical decision support.
Core Techniques: Quantum-Enhanced Raman Spectroscopy, Non-Invasive Optical Glucose Sensing, Wearable Physiological Monitoring, ECG, PPG, EDA, CGM-Based Validation, Photon-Level and Spectral Feature Extraction, AI/ML-Based Glucose Estimation, Hybrid Quantum–Classical Machine Learning, Multimodal Fusion, Signal Denoising, Calibration Modeling, Explainable AI, and Human-Centered Smart Healthcare Evaluation.

2025 – Present **AI & Cybersecurity Red-Teaming Lab**
Focus: AI and mobile device security through adversarial red-teaming, jailbreak analysis, and Deepfake attack simulation.
Outcomes: Initiated the design of an AI & Cybersecurity Red-Teaming Lab to investigate jailbreak vulnerabilities in large language models, iOS devices, and federated learning environments. Developed experimental frameworks for adversarial prompt injection, malicious client simulations, and Deepfake-based attacks on medical imaging systems. Proposed layered defenses, including guardrails for LLMs, forensic imaging verification, and federated anomaly detection, to build resilience against emerging threats.
Core Techniques: Red-Teaming of AI Systems (LLM jailbreak attacks, HarmBench or JailbreakBench testing), Mobile Security (iOS jailbreak tools, kernel exploit analysis), Federated Learning Security (malicious client simulation, gradient poisoning), Deepfake Video Generation (Cross-Attention Guided Identity-Conditional Diffusion, GANs), Quantum-Ready Computing Infrastructure (multi-GPU adversarial training).

2024 – Present **Smart Agriculture: IoT-based Irrigation and Nutrient Management**
Focus: IoT-driven prediction and management of irrigation and nutrient use for sustainable agriculture.
Outcomes: Designed an integrated framework that combines IoT sensors, drones, and machine learning models to optimize irrigation scheduling and nutrient application.

Developed approaches that merge evapotranspiration (ET)-based calculations with soil sensor data, enabling real-time decision-making. Proposed hybrid solar–wind energy harvesting for IoT devices and blockchain protocols for farm data integrity. The system aims to increase crop yield by 30–50%, reduce water and fertilizer waste, and ensure scalability across diverse farm settings.
Core Techniques: IoT Sensing (soil moisture, weather, crop growth stages), Machine Learning (ET modeling, predictive analytics, reinforcement learning), Aerial Imaging (drone-based crop health monitoring), Hybrid Energy Systems (solar + wind harvesting), Blockchain for Data Security, Decision Support Systems, Precision Agriculture.

2020 – Present **Deepfake Analysis, Detection, and Dataset Generation**
Focus: Research on the analysis, detection, and benchmarking of Deepfakes (image/video manipulation), including the development of challenging datasets to advance Deepfake detection research.
Outcomes: Developed pioneering approaches for Deepfake detection using both deep learning and classical machine learning techniques. Introduced a novel framework that combines image processing, computer vision, and statistical feature extraction/selection methods to improve detection performance. Designed and initiated the creation of a custom Deepfake dataset containing authentic and manipulated videos generated using FaceSwap, GAN, and diffusion-based techniques. Applied diverse video compression strategies to simulate real-world social media and online sharing conditions, making detection significantly more challenging. Conducted extensive benchmarking against state-of-the-art Deepfake detection models and performed comprehensive statistical hypothesis testing to evaluate model effectiveness and robustness. The project contributes benchmark resources and analytical methodologies to strengthen cybersecurity research, media forensics, and the protection of online media integrity.
Core Techniques: Deepfake Generation (FaceSwap, GANs, Diffusion Models), Image Processing, Computer Vision, Machine Learning, Deep Learning, Video Processing and Compression, Dataset Design and Benchmarking, Media Forensics, Cybersecurity Applications, Statistical Analysis and Hypothesis Testing (ANOVA, McNemar Test, Cochran's Q Test, F-test, Dietterich's 5x2cv Paired t-Test, Alpaydin's Combined 5x2cv F-test, Nested Cross-Validation, etc.).

2017 – 2019 **Android Malware Detection**
Focus: Detection and prevention of Android Malware.
Outcomes: Proposed a new feature selection method and a Blockchain approach and used various machine learning algorithms to detect Android malware.
Core Techniques: Machine Learning, Statistics (especially Principal Component Analysis), and Blockchain.

2009 – 2015 **Image Inpainting (Restoration) and Medical Image Processing**
Focus: Image inpainting and restoration of degraded images, including filtering and restoring Ultrasound images
Outcomes: Proposed a new image inpainting method for restoring degraded images; A new image filtering method based on the Perona-Malik equation
Core Techniques: Image Processing (image restoration), Mathematics/Numerical Analysis (i.e., Partial Differential Equations, Method of Fundamental Solutions, Convolution theorem, Radial Basis Functions, etc.), Computer Vision, Mathematics/Numerical Analysis (i.e., Partial Differential Equations, Fourier Transform, etc.), Statistics

FUNDS/GRANTS/FELLOWSHIPS INCLUDING PENDING

RESEARCH GRANTS

PENDING

2027 – 2028 **NIH (Limited Competition: Instrumentation Grant Program for Resource-Limited Institutions (S10 - Clinical Trial Not Allowed)): Integrated Raman–NIR Biomedical Optical Sensing System, ~$104k**

Working as PI on a collaborative instrumentation proposal to establish an Integrated Raman –NIR Biomedical Optical Sensing System for non-clinical glucose sensing, tissue-phantom validation, COMSOL digital-twin modeling, AI-enabled calibration, secure data workflows, and student training

2026 – 2029 **NIJ: Shielding Seniors: Real-Time Multimodal Fraud Detection and Agentic Recommendation System for Protecting Older Adults from AI-Enabled Financial Exploitation, ~$375k**

PI on a collaborative project focused on protecting seniors from AI-based financial exploitation, including deepfakes, voice cloning, phishing emails, and fraudulent text messages, while also providing training and raising awareness about emerging threats.

REJECTED

2026 – 2028 **NSF Future CoRe: Collaborative Research: Privacy-Preserving Framework for Healthcare Recommendation with Improved Attack Detection and Mitigation, ~$850k (GSU: $273k)**

Working as a PI in the multi-institutional collaborative project: AI-based Smart Healthcare System using Quantum Sensing & Wearables, Machine Unlearning, and Federated Learning

2026 – 2028 **Research and Education Grants: AI-Driven Multi-Modal Sensor and UAV System for Sustainable Irrigation and Nutrient Management in Smallholder Farming, $400k**

Serving as PI on a collaborative project to improve farming efficiency and enhance economic and environmental sustainability by developing a multi-modal sensor-UAV and AI-based system.

2026 – 2027 **Amazon Academic Research: NeuroVisual-Q-Agent: A Quantum-Augmented Sensing-Based Medical Agent Platform for Wearable Sensing, $100k & AWS Credits (~$70k)**

Working as a PI on the project: Quantum-Augmented Thought-Aware Health Coaching with Wearable Sensing and Just-in-Time Adaptive Interventions

2026 – 2027 **Pathfinder AI: Bridging Skills, Opportunity, and Economic Mobility through Human-Centered AI, $250k** [**Concept Note**]

Working as a PI on a project: a human-centered career mobility platform to help low-income and under-credentialed individuals identify practical, higher-paying career opportunities

2026 - 2030 **NSF CAREER: Securing the Digital Frontier: Detection and Forensic Analysis of Generative AI Content, $522k [Under Resubmission]**
Working as a PI on the project: AI-based content analysis and detection for Enhanced Cyber-Physical Systems Security.

2025 - 2028 **NSF MRI: Major Research Instrumentation Program (MRI), $806k**
Working as a Co-PI on the project: Acquisition of High-Performance Computing Infrastructures for Southwest Florida.

UNDER SUBMISSION/PROCESSING

2027 – 2028 **NSF ERI: Transferable Cross-Modal Signal Relationships for Audiovisual Authenticity under Generative and Signal-Condition Shifts, ~$200k**

Serving as PI, leading research on transferable acoustic–visual signal relationships, cross-generator generalization, perturbation robustness, and reliability-aware audiovisual authenticity assessment.

2027 – 2029 **NSF-IUSE: RAISE-Cyber: A Responsible AI Ecosystem for Strengthening Undergraduate Cybersecurity Pathways, Student Success, and Workforce Readiness, ~$492k**

Working as a Co-PI on a collaborative NSF-IUSE project focused on strengthening undergraduate cybersecurity education through responsible AI, AI-enabled cybersecurity modules, quantum/wearable technology foundations, secure data analytics, and workforce-readiness pathways, while also supporting curriculum development, faculty training, student mentoring, and assessment of learning, confidence, retention, and career preparation.

2027 – 2027 **DoW SBIR: Application of Agentic AI for Mission Critical Implementation of Autonomous Defect Detection in Extrusion-Based Manufacturing, ~$200k**

Serving as Co-PI, leading physics-informed AI and digital-twin development for volumetric manufacturing process optimization, cross-site transferability, and RF/thermomechanical performance validation of low-loss 3D RF interposers.

2026 – 2029 **NSF/NIH (SCH): Collaborative Research: AI-Enabled Wearable Quantum-Inspired Sensing for Non-Invasive Continuous Glucose Monitoring: Toward Intelligent Physiological Monitoring, ~$1.2M**

Serving as PI on a project developing an AI-enabled wearable sensing platform for non-invasive continuous glucose monitoring and next-generation intelligent physiological monitoring systems.

2026 – 2029 **NSF EAGER: Collaborative Research: EAGER: Quantum-Inspired Signal Processing for Wearable Raman Spectroscopy in Continuous Non-Invasive Glucose Monitoring, ~$400k**

Serving as PI on a project developing an AI-enabled wearable sensing platform for non-invasive continuous glucose monitoring and next-generation intelligent physiological monitoring systems.

FELLOWSHIP

2024 – 2025 **Daveler Faculty Fellows, FGCU, $15k**
Innovative Cybersecurity Ventures: Developing Real-Time Threat Detection Solutions.

RESEARCH FUNDS

2026 – 2027 **Strategic Fund, GSU, ~$35k**
For the high-performance computing system for my Red Teaming Lab

2026 – 2027 **Research Seed Funding Awards, GSU, ~$12.5k**
Working as a co-PI in the Cyber Risk Assessment and Mitigation for Drinking Water Systems

2024 - 2025 **Head-Start Program, U.A. Whitaker College of Engineering, FGCU, $5k**
Working as a PI in the Development of a real-time Deepfake Detection application project.

2023 - 2024 **WiSER Eagles (Work-study in Scholarly Experiential Research), FGCU, $15k**
Working as a PI in the Analyzing and Detecting Cybersecurity Threat project

2023 - 2024 **Head-Start Program, U.A. Whitaker College of Engineering, FGCU, $5k**
Working as a PI in the Enhancing Digital Authenticity: Advancing Multimodal Deepfake Detection project.

2023 - 2024 **Head-Start Program, U.A. Whitaker College of Engineering, FGCU, $5k**
Working as a co-PI in the *EagleCyberHub*: Developing a Multidisciplinary Cybersecurity Community of Practice in Southwest Florida project.

Travel Grants

2026 - 2027 **International Research & Engagement Travel Award, GSU, $6k**
Support for (i) establishing research collaboration; (ii) providing a keynote speech and presenting research at the international conference.

2026 **Competitive Faculty Travel Award, GSU, ~$1.5k**
Support for conference registration, attending, or presenting papers.

2023 **Professional Development Fund from the Lucas Center, FGCU, ~$650**
Support for conference registration, attending, or presenting papers.

2023 **Mini-Grants for Travel and Research, FGCU, $1k**
Support for conference registration, attending, or presenting papers.

Selected Participant in Research Workshop

2025 – 2026 NSF CISE RE Aspiring PI Workshop, National Science Foundation (NSF) – CISE Directorate, Miami, FL & Virtual | Dec 2025 – Jan 2026

Competitively selected to attend a prestigious NSF-funded workshop aimed at developing future Principal Investigators (PIs) in Computing and Information Science and Engineering (CISE), a national workshop focused on research leadership, grant writing, and collaboration with NSF program directors and mentors

Supporting RA on DOD TKX Project

2017 – 2019 **Technology Knowledge Exchange (TKX), $5 Million**
Having worked as a research assistant in this project funded by the US Department of Defense (DoD), $5 million total, including a subcontract to the USM industrial partner.

Research Scholars/Interns/Students Supervision

Graduate Students (PhD Dissertation)

2026–Present Tanoy Debnath, AI and Cybersecurity in mitigating Computer Vision Problems, Georgia Southern University

2025–Present Md Anisur Rahman, AI and Cybersecurity in Smart Healthcare, Smart Agriculture, Georgia Southern University

2025- 2025 Md Iqba Hossain, AI and Cybersecurity in mitigating Computer Vision Problems, University of Massachusetts Dartmouth

Graduate Students (MSc Thesis/Projects)

2026–Present Samuel Adeniyi, AI and Cybersecurity in mitigating Computer Vision Problems, Georgia Southern University

2025- 2025 Mohammad Muid Uddin Chowdhury, AI and Cybersecurity in mitigating Mis- and Disinformation Analysis and Detection, University of Massachusetts Dartmouth

2021- 2022 Deep Bhakta, Android Malware Detection, BUP

Undergraduate Research Scholars

2023-2025 Aidan Premeau, Advancing Multimodal Deepfake Detection, FGCU

2022-2023 Jaylen Jones, Analyzing Multimodal Datasets for Detecting Online COVID Misinformation,

Undergraduate Students (BSc Thesis/Project)

2016- 2017 Md. Minul Alam Shikdar and Nazmul Hasan, A Strategic Approach for Tactile Based Braille System, DIU
2016- 2017 Mahfuzur Rahman, E-learning/Distance Learning Application Development, DIU
2014- 2015 Taukir Hasan, Travel & Tourism Automation, DIU
2014- 2015 Md. Sazzad Hossain, Online Newspaper, DIU

Mentoring Undergraduate Students

2022 - 2024 Md. Solaiman, and Md. Fahimuzzaman Sohan, Creating a Robust and Diverse Deepfake Dataset, Bangladesh (Remote)
2016- 2017 Zaman Wahid and Syeda Sumbul Hossain, E-learning/Distance Learning, DIU

Research Profile

Scholar https://scholar.google.com/citations?user=jjM9zsAAAAAJ&hl=en

ResearchGate https://www.researchgate.net/profile/Md-Rana-35

ORCID Md. Shohel Rana (0000-0001-6626-4189) (orcid.org)

Professional Affiliations and Services

Committee Member

2023 – 2025 Course Curriculum Committee, FGCU

2022 – 2023 ABET Committee, University of Alabama at Birmingham (UAB)

2022 – 2023 Undergraduate Curriculum Committee, CS, UAB

2021 – 2022 Advanced Software Engineering Program, Northern Kentucky University (2021-2022)

2017 – 2017 Communication Secretary & Organizer: DIU APP CONTEST 2017

2016 – 2017 Research Committee: Daffodil International University (2016-2017)

Ad-hoc Reviewer

- ✓ IEEE Access
- ✓ IEEE Transactions on Pattern Analysis and Machine Intelligence
- ✓ Journal of Engineering Applications of Artificial Intelligence
- ✓ Journal of KSII Transactions on Internet and Information Systems
- ✓ IEEE International Workshop on Adapting Data Mining for Security 2018.

Technical/Program Committee Member

- ✓ ACM Workshop on Secure and Trustworthy Cyber-Physical Systems, 2023, 2024.
- ✓ International Conference on Information and Computer Technologies (ICICT), 2024, 2025

Professional Member

- ✓ DENDRITIC: A Human-Centered AI and Data Science Institute, FGCU
- ✓ ACM
- ✓ IEEE and IEEE Computer Society
- ✓ Bangladesh Computer Society (BCS)
- ✓ Cyber Security Centre, Daffodil International University

INDUSTRY AFFILIATION

CYBERETTE, a corporation organized and existing under the laws of the Netherlands, with its principal place of business at Gevers Deynootweg, 2586BV, The Hague, The Netherlands. The Company is engaged in the development of cybersecurity solutions, specifically a "**Deepfake Detector**" designed for commercial use.

PUBLICATIONS

PEER-REVIEWED JOURNAL ARTICLES AND INVITED BOOK CHAPTERS

Nobi, M. N., **Rana, M. S.**, and Krishnan, R. (2026). Optimizing Machine Learning-Based Access Control Administration Through Data Distillation. *In: Hubballi, N., Verma, R., Rudrapatna, K., S. (eds) Information Systems Security. ICISS 2025. Lecture Notes in Computer Science, vol 16380*. Springer, Cham. DOI: 10.1007/978-3-032-13714-2_2

Rana, M. S., Ghosh, T., Nobi, M. N., Rahman, A., and Sung, A. H. (2025). Unveiling Zero-Click Attacks: Mapping MITRE ATT&CK Framework for Enhanced Cybersecurity," *Computers, Materials & Continua*, 2025, DOI: 10.32604/cmc.2025.069212.

Rana, M. S., Nurnobi, M., Murali, B., and Sung, A. H. (2022). Deepfake Detection: A Systematic Literature Review, *IEEE Access*.

Bhakta, D., Yousuf, M. A., and **Rana, M. S.** (2022). Android Malware Detection Against String Encryption Based Obfuscation. *Third Congress on Intelligent Systems, CIS 2022*. Lecture Notes in Networks and Systems, Vol. 613. Springer, Singapore.

Rana, M. S. and Sung, A. H. (2020). Evaluation of Advanced Ensemble Learning Techniques for Android Malware Detection. *Vietnam Journal of Computer Science*. Vol. 07, No. 02.

Rana, M. S., Gudla, C., and Sung, A. H. (2019). Evaluating Machine Learning Models on the Ethereum Blockchain for Android Malware Detection. *Intelligent Computing - Proceedings of the Computing Conference*.

Rana, M. S., Rahman, S. S. M. M. and Sung, A. H. (2018). Evaluation of Tree-Based Machine Learning Classifiers for Android Malware Detection. *10th International Conference, ICCCI 2018, Bristol, UK, Proceedings, Part II*.

Rana, M. S., Bhuiyan, T., and Satter, A. K. M. Z. (2018). e-School: Design and Implementation Web Based Teaching Institution for Enhancing E-Learning Experiences. *10th International Conference, ICCCI 2018*, Bristol, UK, Proceedings, Part I.

Rana, M. S., and Sung, A. H. (2018). Malware Analysis on Android using Supervised Machine Learning Techniques. *International Journal of Computer and Communication Engineering*, ISSN: 2010-3743.

Rana, M. S., Sohel, M. K., and Arman, M. S. (2018). Distributed Database Problems, Approaches and Solutions – A Study. *International Journal of Machine Learning and Computing.*

Rana, M. S., Hassan, M. M., and Bhuiyan, T. (2017). An Enhanced Model for Inpainting on Digital Images Using Dynamic Masking. *Journal of Communications*, Vol. 12, No. 4, pp. 248-253.

Rahman, M. M., **Rana, M. S.**, Islam, M. A., Rahman, M. M., and Talukder, M. H. (2012). A New Filtering Technique for denoising Speckle Noise from Medical Images Based on Adaptive and Anisotropic Diffusion Filter. *International Journal of Research in Computer and Communication Technology*, Vol. 2, Issue 9. [**IF: 3.751**]

CONFERENCE PAPERS, POSTERS AND PRESENTATIONS

Rahman, A., Debnath, T., and **Rana, M. S.** (2026). Wearable Intelligent System for Non-Invasive Glucose Monitoring Using Quantum Sensing, *The 17th ACM Conference on Bioinformatics, Computational Biology and Health Informatics (ACMBCB 2026), June 30 – July 3, 2026*, Calabria, Italy. DOI: 10.1145/3807503.3816890

Rana, M. S., Rahman, A., Debnath, T., and Sung, A. H. (2026). QSen-FMoE: A Federated Mixture-of-Experts Framework for Quantum-Driven Next-Generation Wearable Sensing in Body Area Networks. *14th International Conference on Bioinformatics and Computational Biology (ICBCB 2026), March 26-29, 2026*, Kitakyushu, Japan. DOI: 10.1109/ICBCB69424.2026.11621213.

Rahman, A., Walee, N. A., **Rana, M. S.**, Chen, L., and Andrew H. Sung (2026). Privacy-Preserving Explainable Federated Learning for Non-Invasive Blood Glucose Estimation from Photoplethysmography Signals. *18th International Conference on Bioinformatics and Biomedical Technology (ICBBT 2026), May 22-25, 2026*, Suzhou, China. DOI: 10.1145/3822521.3822536. **[In press]**

Coffie, L., Ajuluchukwu, M., Mamud, A., and **Rana, M. S.** (2026). Agentic Mixture-of-Experts Forensics for Detecting and Localizing Deepfake Manipulations in Volumetric Lung Computed Tomography. *2026 7th International Conference on Artificial Intelligence, Robotics, and Control (AIRC), Savannah, GA, USA, 2026, pp. 348-353.* DOI: 10.1109/AIRC69745.2026.11631462.

Herman, K., Herwig, D., and **Rana, M. S.** (2026). Real-Time Deepfake Threat Detection: Benchmarking Lightweight Audio Forensics Against SOTA Video and Hybrid Architectures. *2026 7th International Conference on Artificial Intelligence, Robotics, and Control (AIRC), Savannah, GA, USA, 2026, pp. 361-365.* DOI: 10.1109/AIRC69745.2026.11631362.

Rahman, A., Walee, N. A., and **Rana, M. S.** (2026). A Lightweight CPU-Optimized Web Application for Real-Time Deepfake Detection using Deep Learning. *2026 7th International Conference on Artificial Intelligence, Robotics, and Control (AIRC), Savannah, GA, USA, 2026, pp. 391-396.* DOI: 10.1109/AIRC69745.2026.11631402.

Rahman, A., Walee, N. A., **Rana, M. S.**, and Chen, L. (2026). BIoMT-VFL: Ensuring Blockchain-based Security and Privacy in Internet of Medical Things through Vertical Federated Learning. *2026 7th International Conference on Artificial Intelligence, Robotics, and Control (AIRC), Savannah, GA, USA, 2026, pp. 428-433*. DOI: 10.1109/AIRC69745.2026.11631456

Rana, M. S., Kader, M. A., Rahman, A., Rahman, M. K., and Ahmed, H. (2026). AI-Enabled Soil Health Monitoring Using UAV Imagery and Ground Sensors for Sustainable Irrigation, *5th AI in Agriculture Conference, March 31– April 2, 2026, North Carolina State University*. **[In press]**

Debnath, T., Rahman, A., and **Rana, M. S.** (2026). CA-IDD (Cross-Attention Guided Identity-Conditional Diffusion): A Robust and Controllable Framework for Realistic Face Swapping, 2026 AEP CEC Student Research Symposium, April 24, 2026, Georgia Southern University, Georgia, USA.

Rahman, A., Debnath, T., and **Rana, M. S.** (2026). Quantum-Enhanced Noninvasive Approach for Continuous Glucose Monitoring, 2026 AEP CEC Student Research Symposium, April 24, 2026, Georgia Southern University, Georgia, USA.

Rahman, A., Debnath, T., Sadik, S., **Rana, M. S.**, and Chen, L. (2026). Network and Identity-Centric Forensics for Insider Threat Detection and Reconstruction in Zero-Trust Healthcare Architectures, *IEEE International Conference on E-health Networking, Application & Services (IEEE Healthcom 26), New York, USA, October 19 - 21 2026.* [**In press**]

Debnath, T., **Rana, M. S.**, and Sung, A. H. (2026). QIFSXAI: A Quantum-Inspired Multi-Objective Framework for Compact and Explainable Clinical Feature Selection. *2nd International Conference on Computational Intelligence and Soft Computing (CISCom 2026), University of Malta, Malta, Europe, October 14-16, 2026.* [**In press**]

Chaudhury, T. S., Shams, M. A., Oyshe, K. U., Masud, S. M. R. A., Debnath, T., and **Rana, M. S.** (2026). Predicting Fetal Health with FetalAIX: A Precision Driven Hybrid ANN–LGBM Model with LIME-Based Explainability. *2nd International Conference on Computational Intelligence and Soft Computing (CISCom 2026), University of Malta, Malta, Europe, October 14-16, 2026.* [**In press**]

Rana, M. S., Hasan, M. M., Ahmed, H. and Sung, A. H. (2026). COMSOL Digital Twin For Raman-NIR Non-Invasive Glucose Sensing: From Prototype To Validation, *COMSOL Conference 2026 Boston, October 7-9, Boston, USA*. [**In press**]

Prasad, R., Challa, S. K. R., Ahmed, F., **Rana, M. S.**, and Gudla, C. (2026). MACDS: A Multi-Agent Autonomous Cyber Defense System for Proactive Threat Detection and Response in Cloud Environments. *39th International Conference on Computer Applications in Industry and Engineering (CAINE 2026), San Francisco, CA, USA, October 19-20, 2026*. [**In press**]

Rahman, A., Aishi, A. A., and **Rana, M. S.** (2026). AI-Powered Quantum Sensing for Non-Invasive Diabetes Monitoring System, GS4 Student Scholars Symposium. 221. Link: https://digitalcommons.georgiasouthern.edu/research_symposium/2026/2026/221.

Rahman, A., Aishi, A. A., and **Rana, M. S.** (2026). Leveraging Agentic and Generative AI for Intelligent Global Market Analysis, Georgia Southern Student Scholars Symposium (GS4).

Hasan, M. M., Gudla, C., Sung, A., and **Rana, M. S.** (2026). Zero-Click Attacks and Victims: A Comprehensive Data-Driven Analysis of Targeting Patterns and Exploitation Trends, *35th International Conference in Software and Data Engineering (SEDE 2026), San Francisco, CA, USA, October 19-20, 2026*. [**Accepted**]

Rahman, A., Sadik, S., Shahariar, M., Debnath, T., Gudla, C., Sung, A., and **Rana, M. S.** (2026). Toward Quantum-Enhanced Deepfake Detection: A Hybrid CNN–QSVM Framework with Simulation-based Evaluation on Consumer Hardware, *35th International Conference in Software and Data Engineering (SEDE 2026), San Francisco, CA, USA, October 19-20, 2026*. [**Accepted**]

Rana, M. S., Hasan M. M., and Sung A. H. (2025). Towards Deepfake Detection: An Exploratory Data Analysis for Uncovering Domain-Specific Insights. *2025 International Symposium on Multimedia (ISM 2025), Dec 8–10, 2025*, Naples, Italy. **[In press]**

Rahman, A., and **Rana, M. S.** (2025). QSen-FMoE: A Federated Mixture of Experts Framework for Quantum-Driven Sensing and Wearable Intelligence in Body Area Networks, *MCG-GS Savannah Community Engaged Research Conference*, Monday, November 17, 2025, Georgia, USA.

Khan, H., Ahmed, R., Rahman, M. K., **Rana, M. S.**, and Ahmed, H. (2025). Stiffness-Driven Design of Membrane Sensors for Broadband and Selective Frequency Sensing, *MCG-GS Savannah Community Engaged Research Conference*, Monday, November 17, 2025, Georgia, USA

Ghosh, T., **Rana, M. S.,** Hasan, M. M., Ahmed, F., and Gudla, C. (2025). Robust Password Security: A Method in Preventing Guessing Attacks and Authenticating Users. *2025 8th International Conference on Information and Computer Technologies (ICICT)*, Hawaii-Hilo, HI, USA, 2025, pp. 49-54, DOI: 10.1109/ICICT64582.2025.00014.

Ghosh, T., **Rana, M. S.,** Ahmed, F., Hasan, M. M., and Gudla, C. (2025). Adaptive Password Policies: A New Paradigm for Cyber Resilience in Modern Threat Landscape, *2025 8th International Conference on Information and Computer Technologies (ICICT)*, Hawaii-Hilo, HI, USA, 2025, pp. 55-60, DOI: 10.1109/ICICT64582.2025.00015.

Nobi, M. N., Gupta, M., Krishnan, R., **Rana, M. S.,** Praharaj, L., and Abdelsalam, M. (2025). Machine Learning in Access Control: A Taxonomy [Systematization of Knowledge Paper]. *In Proceedings of the 30th ACM Symposium on Access Control Models and Technologies (SACMAT '25)*. Association for Computing Machinery, New York, NY, USA, 145–156. DOI: 10.1145/3734436.3734453.

Vora, H. H. K., Chowdhury, M. M. U., and **Rana, M. S.** (2025). TWIN: Advancing Digital Trust Through Explainable AI and Multimodal Misinformation Detection. Research Poster. *2025 Sigma Xi Research Exhibition (Sigma Xi 2025)*, University of Massachusetts Dartmouth, MA, USA, April 16-17, 2025.

Sankuru, M., and **Rana, M. S.** (2025). DGM-CAM: Dynamic Gradient Masking with Conflict-Aware Merging for Lifelong Editing of Large Language Models. Research Poster, *2025 Sigma Xi Research Exhibition (Sigma Xi 2025)*, University of Massachusetts Dartmouth, MA, USA, April 16-17, 2025.

Rana, M. S., Nurnobi, M., and Sung, A. H. (2024). DEEPDISTAL: Deepfake Dataset Distillation using Active Learning. *The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024) Workshops, 2024, pp. 7723-7730.*

Rana, M. S., Solaiman, M., Gudla, C., and Sohan, M. F. (2024). Deepfakes – Reality Under Threat? *IEEE 14th Annual Computing and Communication Workshop and Conference (CCWC), Las Vegas, NV, USA*, pp. 0721-0727, DOI: 10.1109/CCWC60891.2024.10427659.

Rana, M. S., and Sung, A. H. (2024). Advanced Deepfake Detection using Machine Learning Algorithms: A Statistical Analysis and Performance Comparison. *IEEE 7th International Conference on Information and Computer Technologies (ICICT), Honolulu, HI, USA, 2024, pp. 75-81*. DOI: 10.1109/ICICT62343.2024.00019.

Rana, M. S., Gudla, C., and Nurnobi, M. (2024). Utilizing Machine Learning to Enhance Infrastructure Resilience in Cold Regions. *20th International Conference on Cold Regions Engineering, May 13 – 16, 2024, Anchorage, AK, USA. pp. 12–23*, DOI: 10.1061/9780784485460.002.

Solaiman, M. and **Rana, M. S.** (2024). Enhancing Global Security: A Robust CNN Model for Deepfake Video Detection. *IEEE 7th International Conference on Information and Computer Technologies (ICICT), Honolulu, HI, USA, 2024, pp. 238-243.* DOI: 10.1109/ICICT62343.2024.00044.

Dilworth, R., Gudla, C., **Rana, M. S.**, and Feroz Ahmed (2024). A Disambiguation of Security-based Software Testing. *IEEE 7th International Conference on Information and Computer Technologies (ICICT), Honolulu, HI, USA, 2024, pp. 40-45.* DOI: 10.1109/ICICT62343.2024.00013.

Nampalli, V. V., Gudla, C., and **Rana, M. S.**, (2024). Predicting Travel Time in Complex Road Structures using Deep Learning. *IEEE 14th Annual Computing and Communication Workshop and Conference (CCWC), Las Vegas, NV, USA, pp. 0674-0681,* DOI: 10.1109/CCWC60891.2024.10427657.

Chattopadhyay, A., Beyene, N., and **Rana, S.** (2023). Analyzing Multimodal Datasets for Detecting Online COVID Misinformation: A Preliminary Survey Study*," IEEE 13th Annual Computing and Communication Workshop and Conference (CCWC), Las Vegas, NV, USA, 2023, pp. 0147-0153*.

Rana, M. S., Murali, B. and Sung, A. H. (2021). Deepfake Detection Using Machine Learning Algorithms. *2021 10th International Congress on Advanced Applied Informatics (IIAI-AAI), Japan, pp. 458-463.* DOI: 10.1109/IIAI-AAI53430.2021.00079.

Rana, M. S., and Sung, A. H. (2020). DeepfakeStack: A Deep Ensemble-based Learning Technique for Deepfake Detection. *7th IEEE International Conference on Cyber Security and Cloud Computing (CSCloud 2020)*, New York, USA.

Rana, M. S., Hossin, A., Mahmud, S. M. H., Jahan, H., Satter, A. K. M. Z., and Bhuiyan, T. (2019). MinFinder: A New Approach in Sorting Algorithm. *9th International Conference of Information and Communication Technology (ICICT-2019)*, Nanning, Guangxi, China.

Rana, M. S., Gudla, C., and Sung, A. H. (2018). Evaluating Machine Learning Models for Android Malware Detection: A Comparison Study. *In Proceedings of the 2018 VII International Conference on Network, Communication and Computing (ICNCC 2018)*. Association for Computing Machinery, New York, NY, USA.

Rana, M. S., Hossin, M. A., Mahmud, S. M. H., and Jahan, H. (2018). A new method to handle Facebook users in the distributed database system. *9th IEEE International Conference on Software Engineering and Service Science (ICSESS 2018)*, Beijing, China.

Rana, M. S., Gudla, C., and Sung, A. H. (2018). Android Malware Detection using Stacked Generalization, *27th International Conference on Software Engineering and Data Engineering (SEDE 2018)*, New Orleans, United States.

Gudla, C., **Rana, M. S.,** and Sung, A. H. (2018). Defense Techniques Against Cyber Attacks on Unmanned Aerial Vehicles, *16th International Conference on Embedded Systems, Cyber-physical Systems and Applications*, Las Vegas, United States.

Rana, M. S., Satter, A. K. M. Z., Wahid, Z., and Bhuiyan, T. (2017). Inpainting on Digital Images using Convolution based Method - A Comparison Study, *International Conference on Biomedical Engineering and Bioinformatics*, Bangkok, Thailand.

Bhuiyan, T., **Rana, M. S.**, Sarker, K. and Wahid, Z. (2017). A Proposed PST Model for Enhancing E-Learning Experiences. *International Conference on Education and Distance Learning*, Maldives.

Rana, M. S., Sarker, K. Bhuiyan, T., and Hassan, M. M. (2017). Comparing the performance of different ultrasonic images enhancement for speckle noise reduction in ultrasound images using techniques: a preference study, *2nd International Workshop on Pattern Recognition*, Singapore.

KEYNOTE SPEAKERS/INVITED TALKS/TUTORIALS

- Deepfake Detection: A Tutorial, *9th ACM International Workshop on Security and Privacy Analytics, Charlotte, North Carolina, April 2023*
- Analysis and Detection of Deepfake, *The 2022 Southeast Symposium on Contemporary Engineering Topics, University of Arkansas at Little Rock, September 2022*
- Deepfakes and Detection Techniques, *Annual Meeting, IEEE Mississippi Section*, 2021
- When Seeing Is No Longer Believing: How AI Is Challenging Truth and How We Shape the Future of Trust, *TEDxSavannah, Fine Arts Auditorium at Georgia Southern's Armstrong Campus, Friday, May 15, 2026.*
- Quantum Sensing and Wearables in Non-invasive Continuous Glucose Monitoring (CGM), *2nd International Conference on Computer Science & Engineering (ICCSE 2026), November 23–25, 2026, Paris, France.*
- The Age of Synthetic Reality: AI, Deepfakes, and the Future of Trust. *1st IEEE International Conference on Engineering and Frontier Technologies (ICEFronT 2026), Mawlana Bhashani Science and Technology University, Bangladesh, June 25-26, 2026, Tangail, Bangladesh*

MANUSCRIPTS UNDER SUBMISSION

Rahman A., **Rana M. S.**, Kundu, D., Mahir, S. H., Tashrif, M. T. A., Karim, K. M. A., Debnath, T., Gudla, C., and Sung, A. H. (2025). Unmasking Vulnerabilities in LLMs: A Deep Dive into Jailbreak Attacks and Countermeasures. Cluster Computing. **[Q1-journal] – [Under Review]**

Debnath, T., Bari, M. M., Rabbi, M. S. H., Annanya, T. W., and **Rana, M. S.** (2026). A Hybrid Ensemble Framework for Chronic Kidney Disease Prediction Integrating Classical Machine Learning and Deep Learning with Advanced Interpretability, Computer Methods and Programs in Biomedicine. **[Q1-Journal] – [Under Review]**

Rahman, A, Sadik, S., Hossain, M. M, Debnath, T., Sung, A., and **Rana, M. S.** (2026). Quantum-Enhanced Deepfake Detection: A Hybrid CNN-QSVM Framework for Consumer Hardware Deployment, *1st IEEE International Conference on Engineering and Frontier Technologies (ICEFronT 2026), June 25–26, 2026*, Tangail, Bangladesh. **[Under Review]**

Rana, M. S., Rahman, A., and Debnath, T. (2026). Next-Generation Quantum-Inspired Wearable Raman Platform for Non-Invasive Continuous Glucose Monitoring (CGM), *34th Conference on Intelligent Systems for Molecular Biology (ISMB 2026), July 12–16, 2026,* Washington, D.C., USA. **[Under Review]**

Rana, M. S., Rahman, A., Debnath, T., and Sung, A. H. (2026). Revolutionizing Smart Healthcare System through Quantum Sensing and Wearables: Recent Advances, Use cases, Applications, Challenges, and Future Perspective. *Journal of Engineering Applications of Artificial Intelligence, Elsevier*. **[Q1-journal] – [Under Submission]**

Rana, M. S., and Sung, A. H. (2026). DYMAPIA: A Multi-Domain Framework for Detecting AI-based Video Manipulation. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026), June 19 – 26, 2026*, Denver, USA. **[Under resubmission] – [World-leading conference in computer vision and pattern recognition]**

Debnath, T., **Rana, M. S.**, and Hossain, M. I. (2026). IRCA-Diffusion: Region-Adaptive Cross-Attention Fine-Tuning for Identity-Consistent Face Swapping. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026), June 19 – 26, 2026*, Denver, USA. **[Under resubmission] – [World-leading conference in computer vision and pattern recognition]**

Hasan, M. M., **Rana, M. S.**, and Sung, A. H. (2025). Zero-Click Attacks and the Victims: A Comprehensive Data-Driven Analysis of Targeting Patterns and Exploitation Trends. *Journal of Discover Applied Sciences, SpringerLink*. **[In progress] – [Q2-journal]**

Rana, M. S., Debnath, T., Rahman, A., Nobi, M. N., Solaiman, M., and Sung, A. H. (2027). VDFD: A Diverse and Versatile Benchmark Dataset for Deepfake Research. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2027)*. **[In progress] – [World-leading conference in computer vision and pattern recognition]**

Rana, M. S. and Sung, A. H. (2026). A Meticulous Analysis of Deepfake Datasets for Uncovering Deceptive Videos. [In progress]

Skills and Professional Development

TECHNICAL SKILLS AND COMPETENCES

COMPUTER SKILLS AND COMPETENCIES

Programming	C, C++, C#, ASP.NET, T-SQL, LINQ, MATLAB, Python, and VHDL.
Database	PostGRE, CouchDB, Cassandra, MySQL, SQL Server, and Oracle.
Services	Web/Windows Service, WebAPI 2.0, REST, SOAP, Bluemix
Data Parsing	JSON, ODATA, XML, ATOM-XML
Design Pattern	Repository, N-injection, Dependency Injection, Decorator, and MVC
Scripting Language	jQuery, JavaScript, and Ajax.
Business Intelligence	PreEmptive Business Solution, New Relic, and Tableau.
Modeling Language	UML (Unified Modeling Language)
Version Control	SVN, Microsoft Visual SourceSafe, TFS
Unified Communication	XMPP
Operating System	Windows, Linux (Ubuntu, CentOS), MAC OSX, and Solaris.
Miscellaneous Tools	Tensorflow, Keras, Scikit-learn, OpenCV, PyTorch, qtebe

ORGANIZATIONAL SKILLS AND COMPETENCES

TRAINING/WORKSHOP

- Project Unlock: Leveraging IABs to Infuse EM (PULIAB), Albuquerque, NM, 2024
- Teaching Effectiveness: Active Learning in the Classroom, UAB, 2022
- Teaching Effectiveness: Using Discussions in Face-to-face Classes, UAB, 2022
- Tweaking Lectures for Optimal Comprehensibility, UAB, 2022
- Understanding the Impact of Cross-cultural Values, UAB, 2022
- Supporting the Development of Critical Thinking Skills, UAB, 2022

Arrange seminars, workshops, and programming contests.

Teaching Experience

Daffodil International University

Spring 2014	Software Engineering Project- IV: Database Programming
Fall 2014	Project/Thesis, Software Engineering Project II (Web Programming)

Spring 2015	Object-Oriented Concept & Design with Lab, Object-Oriented Software Development
Fall 2015	Desktop and Web Programming, Software Engineering Project-1
Spring 2016	Desktop and Web Programming, Project/Thesis
Fall 2016	Project/Thesis, Operating System, Software Project Management, Software Engineering Project-1
Spring 2017	Project/Thesis, Computer Fundamentals with LAB, Dot NET Programming with Lab
Summer 2017	Desktop and Web Programming with LAB, Dot NET Programming with Lab

University of Southern Mississippi (As a TA)

Fall 2017	Computer Systems, Data Structures
Spring 2018	Lab Instructor for master's students
Spring 2020	Computer Science II (C++)
Spring 2021	Computer Science II (C++)
Fall 2020	Computer Science II (C++)

Northern Kentucky University

Fall 2021	Information Technology Fundamentals, Networking Fundamentals, Windows Administration
Spring 2022	Information Technology Fundamentals, Networking Fundamentals

University of Alabama at Birmingham

Summer 2022	Operating System, Algorithms and Data Structures
Fall 2022	Object Oriented Programming, Algorithms and Data Structures, Software Engineering, Introduction to Data Science
Spring 2023	Object Oriented Programming, Algorithms and Data Structures, Capstone Project

Florida Gulf Coast University

Fall 2023	Software Engineering Fundamentals, Data Science & Engineering System Analysis
Spring 2024	Programming I (2 sections)
Summer-A 2024	Programming II
Fall 2024	Design and Analysis of Algorithm, Event-Driven Programming (C#)

University of Massachusetts Dartmouth

Spring 2025	Secure Software Development

Georgia Southern University

Fall 2025	AI in Real-time Cybersecurity
Spring 2026	Network Security
Fall 2026	Network Security, Fundamentals of Information Systems Security

TEACHING INTERESTS

Lectures: Intro to Information Technology, Computer Science I & II, Programming Fundamentals I, II & III, Structured Programming, Object-Oriented Programming (C++/Java), Event Driven Programming (C#), Data Structures and Algorithms, Design and Analysis of Algorithm, Object-Oriented Analysis and Design (UML), Desktop and Web Programming, Software Design and Development, Advanced Web Application Development, Database Programming (PL/SQL), Software Engineering, Digital Electronics, Data Analytics, Data Science, Data Handling & Analysis Techniques, Machine Learning, Computer Vision, Machine Vision (Digital Image Processing), Computer and Network Security, Information Security, Secure Software Development, Digital Media Forensics, Introduction to Cybersecurity, Independent Study, Topics in Computer Science.

Guide: Research project/thesis in Computer Science, in Cybersecurity, Data Science, Machine Learning, Image Processing & Computer Vision.

REFERENCES

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